(vi) Bepaal die maksimum spoed (in km h^{-1}) wat die voertuig om die draai kan gaan.
Determine the maximum speed (in km h^{-1}) that the motorist can take the curve.

$m\frac{v^2}{R} = \mu_s F_N \cos\theta + mg\,\cancel{\cos\theta}\,\sin\theta$

$m\frac{v^2}{R} = \mu_s \left(\frac{mg\,\cancel{\cos\theta}}{\cos\theta}\right).\cos\theta + mg\,\cancel{\cos\theta\,\sin\theta}\,\tan\theta$

$\therefore v = \sqrt{R(\mu_s g \cancel{\cos^2\theta} + g\,\cancel{\cos\theta\,\sin\theta}\,\tan\theta}$

$= \sqrt{320(0{,}8(9{,}8)(\cos^2 5{,}1°) + (9{,}8)(\cos 5{,}1°)(\sin}$ $(\tan 5{,}1°)$

$= 52{,}8$ m/s

$= 190{,}1$ km/h

3

(4)

VRAAG / QUESTION 3

(a) Definieer hoeksnelheid ω en gee sy eenhede.
Define angular velocity ω and give it's units.

It is the rate of change of angular displacement (radian sec

2

(2)

(b) Toon aan dat die verband tussen lineêre snelheid v en hoeksnelheid ω gegee word deur v = ωr.
Show that the relation between linear velocity v and angular velocity ω is given by v = ωr.

$\theta = \frac{s}{r}$ $\quad \omega = \frac{d\theta}{dt} = \frac{d}{dt}\left(\frac{s}{r}\right) = \frac{1}{r}\left(\frac{ds}{dt}\right) = \frac{v}{r}$

$\therefore v = \omega r$

3

(3)

(c) 'n Voorwerp wat met hoeksnelheid ω roteer word opgedeel in deeltjies Δm_i wat 'n lineêre snelheid $\bar{v}_i$ het. Δm_i is op afstand r_i vanaf die rotasie-as. Bepaal die rotasie kinetiese energie en toon aan dat die traagheidsmoment $I = \int r^2 dm$.

An object rotates with angular velocity ω and is devided in small particles Δm_i, each with linear velocity $\bar{v}_i$. Δm_i is at a distancer r_i from the axis of rotation. Determine the rotational kinetic energy and show that the moment of inertia is $I = \int r^2 dm$

Vraag / *Question* 3(c)(Vervolg / *Continue*)/ ....